

Highlights

Query-Guided Graph Neural Networks for Knowledge Graph Reasoning

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- We propose a Query-Conditioned Relational Aggregator (QCRA) that embeds query semantics into message construction, enabling selective and relation-aware information propagation.
- We design an Enhanced Structure Align Fusion (ESAF) module that adaptively aligns and integrates multi-layer representations to preserve hierarchical semantics and improve reasoning robustness.
- The proposed framework achieves strong performance across multiple benchmark datasets, demonstrating its effectiveness and stability.

Query-Guided Graph Neural Networks for Knowledge Graph Reasoning

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Abstract

Knowledge graph reasoning aims to infer missing or implicit facts by capturing the structural and semantic regularities underlying observed triples. While graph neural networks (GNNs) have become a dominant paradigm for this task, most existing approaches fail to explicitly incorporate the semantics of the query relation into the message passing process, leading to noisy and indiscriminate aggregation. Furthermore, despite the potential of multi-layer GNNs to encode hierarchical information, current fusion mechanisms are often static and fail to adaptively exploit complementary semantics across layers. To address these limitations, we propose a novel GNN-based reasoning framework that integrates query-guided message propagation and adaptive hierarchical fusion. Specifically, the Query-Conditioned Relational Aggregator (QCRA) embeds query semantics into both message construction and attention computation to enable selective, relation-aware propagation. Meanwhile, the Enhanced Structure Align Fusion (ESAF) module aligns and adaptively integrates representations across layers, enhancing both semantic expressiveness and structural consistency. Extensive experiments on multiple transductive and inductive benchmarks demonstrate that our model consistently outperforms strong baselines in terms of Mean Reciprocal Rank (MRR) and Hits@K, validating the effectiveness and generalizability of our query-guided, structure-aware reasoning framework.

Keywords:

Knowledge graph, Graph neural network, Knowledge graph reasoning

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1. Introduction

The rapid advancement of artificial intelligence has led to a growing demand for structured knowledge representation. Knowledge graphs (KG) have emerged as an effective paradigm for organizing factual information about entities in the real world and their semantic relationships Chen et al. (2020); Ji et al. (2022). Typically represented as a collection of triples (head, relation, tail), KGs support a wide range of downstream applications Fu et al. (2023), including semantic search Qiu et al. (2017); Xiong et al. (2017a), question answering Li et al. (2024), and recommendation systems Tang et al. (2024). One of the central tasks in this domain is knowledge graph reasoning, which aims to infer missing or implicit facts based on existing triples, thus improving the completeness and utility of the knowledge graph Battaglia et al. (2018); Li et al. (2023).

Traditional approaches to this task have mainly relied on knowledge graph embedding (KGE) methods Dettmers et al. (2018); Vashishth et al. (2020a), which represent entities and relations in continuous vector spaces and perform reasoning through vector space. KGE models such as TransE Bordes et al. (2013), DistMult Yang et al. (2014), and ComplEx Trouillon et al. (2016) have shown strong performance on benchmark datasets. However, because KGE models rely on fixed, context-independent embeddings for all entities and relations, they struggle to adapt to varying query contexts and suffer from poor scalability on large or dynamic knowledge graphs Zhao et al. (2021). To address the limitations of traditional embedding-based methods in the capture of complex relational structures, recent studies have explored graph neural networks (GNN) as a more expressive framework for understanding graph reasoning Liao et al. (2022). Using multi-hop message passing, GNNs enable relational information to be propagated and aggregated across graph neighborhoods, offering improved capacity for modeling structured dependencies Chen et al. (2023); Kipf and Welling (2017).

Although GNNs Liu et al. (2021a) have shown a strong potential for multi-hop reasoning over knowledge graphs Bansal et al. (2019), existing models often fail to effectively incorporate the semantics of the query relation during message propagation. In most cases, the query relation serves only as a passive signal or is introduced at the final scoring stage, leaving neighborhood aggregation largely query-agnostic. This design leads to an indiscriminate message flow, allowing irrelevant neighbors to dominate and diluting the reasoning signal Wang et al. (2021a); Zhang and Yao (2022). For example,

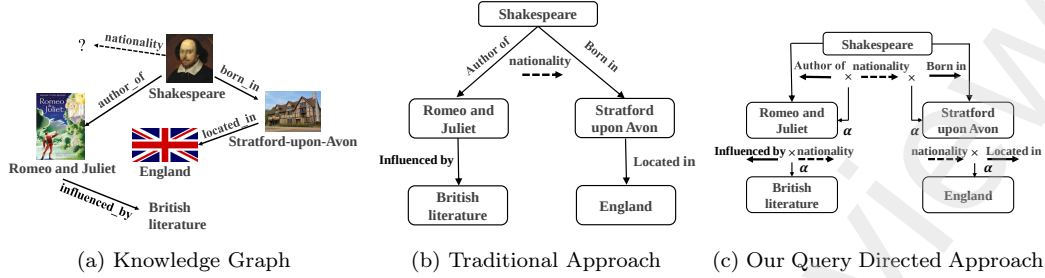


Figure 1: (a) A subgraph centered on ‘Shakespeare’ with entities such as ‘England,’ and relations like ‘author of’, where $\rightarrow$ denotes a relationship and $\dashrightarrow$ denotes a query relationship. (b) Traditional GNN-based reasoning, where messages are propagated uniformly across all neighbors, regardless of the query relation. (c) Our query-directed approach, which injects query relation semantics to modulate message flow via attention α .

consider the query (Shakespeare, born in, ?) shown in Figure 1. Traditional GNNs (Figure 1b) aggregate messages uniformly from all neighbors, giving equal weight to relations like the ones authored or influenced by, which are irrelevant to the query. Such noisy propagation obscures critical paths such as Shakespeare $\rightarrow$ Stratford upon Avon $\rightarrow$ England, which directly answer the query.

Another key limitation lies in hierarchical representation fusion. Although multi-layer GNNs naturally capture different levels of abstraction, ranging from local patterns to global topology, most existing models rely on static or uniform fusion strategies (e.g., GRU-based aggregation Zhang et al. (2023)). These simplistic approaches neglect semantic complementarity across layers, often leading to redundancy or feature dilution, which ultimately hinders reasoning accuracy.

To address these challenges, we propose a novel GNN-based framework for knowledge graph reasoning that performs query-guided message propagation and adaptive hierarchical fusion (Figure 1c). Unlike conventional aggregation methods, our approach explicitly incorporates query semantics into both message construction and propagation, enabling selective information routing along query-relevant paths while suppressing noise. The framework consists of two key components: a Query-Conditioned Relational Aggregator (QCRA), which embeds the query relation into message generation and weighting to dynamically score neighbor importance based on semantic compatibility; and an Enhanced Structure Align Fusion (ESAF) module, which aligns and adaptively integrates node representations across GNN layers,

preserving both shallow and deep semantics to mitigate information loss and enhance robustness in multi-hop reasoning.

In summary, the main contributions of this paper are as follows.

- We proposed a new GNN reasoning architecture built around the QCRA, which enables relation-aware, query-guided message propagation tailored to the semantics of knowledge graph reasoning tasks.
- We designed the ESAF module to align and fuse hierarchical features across GNN layers, enhancing the structural depth and semantic expressiveness of node representations.
- We conducted comprehensive experiments on multiple public benchmarks in both transductive and inductive settings, demonstrating that our method consistently achieves strong performance.

2. Related Work

Knowledge graph reasoning, also known as link prediction, aims to infer missing or implicit facts based on known triples Wang et al. (2021b). It plays a central role in enhancing the completeness and utility of knowledge graphs across various downstream tasks such as question answering, recommendation, and semantic search.

2.1. *Embedding-Based Reasoning Methods*

Early approaches to KG reasoning are dominated by embedding-based models Balazevic et al. (2019); Lin et al. (2015), which learn continuous vector representations for entities and relations, enabling inference through algebraic operations in embedding space. Representative methods include TransE Bordes et al. (2013), DistMult Yang et al. (2014), ComplEx Trouillon et al. (2016), RotatE Sun et al. (2018), and ConvE Dettmers et al. (2018). These models have shown competitive performance on benchmark datasets, but they often rely on shallow relational patterns and lack the ability to model long-range dependencies or path-based reasoning.

2.2. *GNN-Based Reasoning Methods*

To address the structural limitations of embedding methods, recent work has explored the use of Graph Neural Networks (GNNs) for KG reasoning

Chen et al. (2023); Kipf and Welling (2017); Zhao et al. (2021). GNNs operate through message passing, allowing entities to aggregate information from their neighbors over multiple hops. Among them, R-GCN Schlichtkrull et al. (2018) introduces relation-specific aggregation to update entity representations through 1-hop neighbors. CompGCN Vashishth et al. (2020b) extends this framework by jointly modeling entities and relations using compositional operations. These models capture more expressive semantics, but often involve full-graph propagation, which can lead to high memory consumption and over-smoothing.

2.3. *Progressive Propagation Methods*

To improve efficiency, subsequent models adopt progressive propagation Chen et al. (2022); Liu et al. (2021b), where message passing is conducted from the query entity to its multi-hop neighborhood. For example, NBFNet Zhu et al. (2021) models the representation of a target node as the aggregated signal from all paths emanating from the query, while RED-GNN Zhang and Yao (2022) builds a relational digraph and applies recursive attention mechanisms for relational path encoding. These approaches avoid constructing per-query subgraphs, yet still incur computation over large neighborhoods, particularly in deeper layers. Another line of work introduces constrained propagation, such as GraIL Teru et al. (2020) and CoMPiLE Mai et al. (2021), which extract local subgraphs between query–candidate pairs to perform targeted reasoning. Although more precise, these methods suffer from high computational cost, as a separate subgraph must be encoded for each query. To mitigate the inefficiency of full and progressive propagation, attention-based models such as AdaProp Zhang et al. (2023) and A*Net Zhu et al. (2023) dynamically select top- k neighbors or edges at each GNN layer, focusing on the salient paths while discarding noisy or redundant structures. However, such pruning may risk losing important reasoning signals, especially in sparse or noisy graphs.

3. The Proposed Method

In this section, we present the proposed GNN-based framework for knowledge graph reasoning. Figure 2 illustrates the overall architecture of our proposed framework, which dynamically encodes query-specific subgraphs and performs relation-aware reasoning without relying on pre-trained entity embeddings. The framework comprises two key components: (1) the

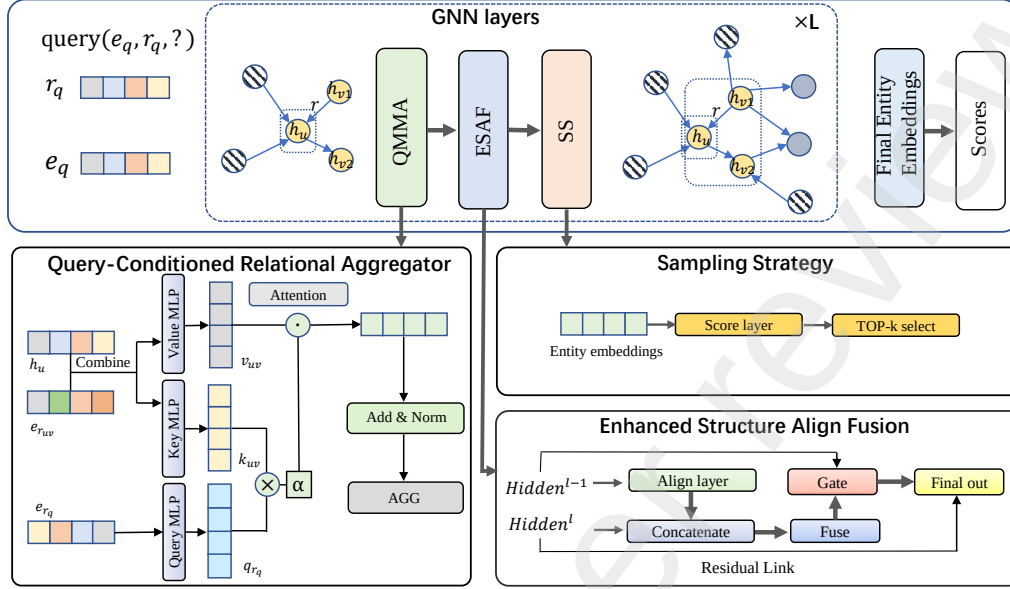


Figure 2: Overall framework of the proposed model. It consists of two key components: (1) Query-Conditioned Relational Aggregator (QCRA) for query-guided message passing, (2) Enhanced Structure Align Fusion (ESAF) for adaptive multi-layer representation fusion.

Query-Conditioned Relational Aggregator (QCRA), which guides message aggregation using query semantics; and (2) the Enhanced Structure Align Fusion (ESAF) module, which adaptively fuses multi-layer representations to capture both local and global structural information.

3.1. Problem Formulation

A knowledge graph (KG) is formally defined as $\mathcal{G} = (\mathcal{V}, \mathcal{R}, \mathcal{T})$, where $\mathcal{V}$ denotes the entity set, $\mathcal{R}$ the relation set, and $\mathcal{T} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$ the collection of factual triples (h, r, t) expressing relational knowledge. The knowledge graph completion (KGC) task addresses incomplete queries $(h, r, ?)$ or $(?, r, t)$ by modeling the conditional probability distribution over candidate entities, wherein during training, a KGC model learns parameters θ to maximize the plausibility scores $f_\theta(h, r, t)$ for all triples in the training set $\mathcal{T}_{train}$; subsequently, at reasoning time, the model ranks candidate head/tail entities for unseen test triples $\mathcal{T}_{test}$ (with $\mathcal{T}_{test} \cap \mathcal{T}_{train} = \emptyset$) and evaluates performance using ranking metrics such as MRR and Hits@ k , fundamentally constituting an entity ranking optimization problem that seeks to position the ground-truth entities at the highest possible ranks.

3.2. Query-Conditioned Relational Aggregator

To overcome the limitation of query-agnostic neighborhood aggregation in existing GNN-based reasoning models, we propose a Query-Conditioned Relational Aggregator (QCRA) that explicitly incorporates query semantics into the message passing process. Unlike traditional methods that treat all neighbors uniformly, QCRA dynamically adjusts message propagation based on the compatibility between the query relation and local graph structures. This design enables the model to focus on semantically relevant paths, thereby improving reasoning accuracy in multi-relational knowledge graphs.

Given a query triple $(e_q, r_q, ?)$, the model expands a local subgraph $\mathcal{G}_q$ centered on the head entity e_q and propagates information layer by layer. Each propagation step consists of three key operations: query-conditioned message construction, compatibility-based attention scoring, and weighted neighborhood aggregation.

3.2.1. Input Representations

Unlike traditional GNN models that assign initial embeddings to all nodes, our framework initializes only the query head entity e_q using the query relation embedding:

$$h_{e_q}^{(0)} = e_{r_q} \in \mathbb{R}^d.$$

Other nodes in the subgraph are introduced progressively during neighborhood expansion and obtain representations through message passing. Each relation type r is associated with a learnable embedding $e_r \in \mathbb{R}^d$, allowing the model to capture semantic dependencies between relations. The query relation embedding e_{r_q} serves as the guiding signal for selective information propagation.

3.2.2. Message Construction

At layer l , the initial step is to construct messages from each neighbor $u \in \mathcal{N}(v)$ of a target node v . Each message integrates the following elements: the current representation of the neighbor node h_u^l , the embedding of the relation $e_{r_{uv}}$ connecting u and v , and the query relation embedding e_{r_q} . We construct a message embedding that reflects structural context and query semantics using a query-key-value formulation:

$$\begin{aligned}
k_{uv}^{(l)} &= W_k \cdot (h_u^{(l)} \odot e_{r_{uv}}) \\
v_{uv}^{(l)} &= W_v \cdot (h_u^{(l)} \odot e_{r_{uv}}) \\
q_{r_q} &= W_q \cdot e_{r_q}
\end{aligned}$$

where $\odot$ denotes element-wise interaction and $W_k, W_v, W_q \in \mathbb{R}^{d \times d}$ are learnable projection matrices. The message key and value incorporate both node and relation information, while the query projects the semantic intent of e_{r_q} into the attention space.

3.2.3. Query-Aware Attention Score Computation

To determine the compatibility between the query and a candidate neighbor, we compute the attention score $\alpha_{uv}^{(l)}$ using a scaled dot-product mechanism:

$$\tilde{\alpha}_{uv}^{(l)} = \frac{(q_{r_q})^\top k_{uv}^{(l)}}{\sqrt{d}},$$

followed by normalization over the neighborhood:

$$\alpha_{uv}^{(l)} = \frac{\exp(\tilde{\alpha}_{uv}^{(l)})}{\sum_{k \in \mathcal{N}(v)} \exp(\tilde{\alpha}_{kv}^{(l)})}.$$

This design ensures that the influence of each neighbor is modulated by its semantic compatibility with the query relation, effectively filtering out irrelevant paths.

3.2.4. Query-Aware Aggregation

The representation of node v in the next layer is computed as a weighted combination of value vectors:

$$h_v^{(l+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} \alpha_{uv}^{(l)} \cdot v_{uv}^{(l)} \right),$$

where $\sigma(\cdot)$ denotes a non-linear activation such as ReLU or GELU.

3.3. Enhanced Structure Align Fusion Module

While GNNs enable multi-hop information aggregation through stacked layers, each layer captures distinct semantic granularity: lower layers encode local relational patterns, whereas higher layers capture more abstract and global topological information. However, most existing approaches fuse these multi-layer representations using sequential gating mechanisms such as GRUs, which, despite their temporal design, fail to explicitly consider structural alignment or semantic complementarity across layers. This often results in information redundancy or misalignment, limiting the expressive power of the learned representations and degrading reasoning performance on complex knowledge graphs.

To address this, we propose the ESAF Module, which is designed to align, weigh, and fuse hierarchical representations in a query and structure-aware manner. The ESAF module ensures that node embeddings from different GNN layers are dynamically integrated based on their semantic contributions, improving representation robustness and expressiveness.

3.3.1. Structure Alignment Network

To make representations from different GNN layers comparable, we first project them into a unified latent space via layer-specific linear transformations: The first step is to project the historical feature h^{l-1} into the same semantic space as the output of the current layer using a two-layer MLP:

$$\tilde{h} = W_2 \cdot \text{ReLU}(W_1 \cdot h^{(l-1)})$$

where W_1 and W_2 are learnable parameters. This alignment operation ensures compatibility between current and historical node representations, which may originate from distinct graph encoders.

3.3.2. Attention-Gated Fusion

Once the features are aligned, the ESAF module constructs a joint representation:

$$x = [\tilde{h} \parallel h^{(l)}] \in \mathbb{R}^{N \times 2d}$$

where $\parallel$ is the connection operation. A fusion gate is then computed to determine the relative importance of the current versus previous layer features for each node:

$$g = \sigma(W_g^{(2)} \cdot \text{ReLU}(W_g^{(1)} \cdot x)) \in \mathbb{R}^{N \times 1}$$

Among them, σ represents the sigmoid function, and $W_g^{(1)}$ and $W_g^{(2)}$ are learnable parameters. The fused output is computed as

$$f = g \cdot \text{Fuse}(x) + (1 - g) \cdot h^{(l-1)}$$

Here, $\text{Fuse}(\cdot)$ is a separate two-layer MLP that learns an enhanced intermediate representation from the concatenated features. The gate $g \in (0, 1)$ adaptively balances current features and fused representations.

3.3.3. Residual Enhancement

To preserve and reinforce original semantic signals, we apply a residual connection with the current-layer output:

$$h_{out}^{(l)} = f + h^{(l)}$$

This module plays a pivotal role in our model's ability to perform fine-grained layer aggregation.

3.4. Final Scoring and Training Objective

After L layers of query-guided message propagation and hierarchical fusion, each entity node obtains a final representation $h^{(L)}$ encoding both structural and relational semantics. For a given query $(e_q, r_q, ?)$, we compute the unnormalized relevance score for each candidate entity i as:

$$s_i = W_f \cdot h_i^{(L)},$$

where $W_f \in \mathbb{R}^{1 \times d}$ is a learnable projection vector shared across entities.

After obtaining all entity scores, the model is trained using a multi-class log-loss objective that encourages higher scores for true triples and suppresses incorrect ones. Given a training set $\mathcal{T} = \{(e_s, r_q, e_o)\}$, the overall loss is formulated as:

$$\mathcal{L} = \sum_{(e_s, r_q, e_o) \in \mathcal{T}} \left(-f(e_s, r_q, e_o) + \log \sum_{e \in \mathcal{E}} \exp(f(e_s, r_q, e)) \right),$$

where $f(e_s, r_q, e)$ denotes the predicted score of entity e under the query $(e_s, r_q, ?)$ computed through the final projection layer.

Minimizing $\mathcal{L}$ enlarges the scores of positive triples while penalizing negatives obtained by replacing the correct object entity with any other entity from $\mathcal{E}$.

3.5. Sampling Strategy for Subgraph Construction

We adopt a sampling strategy commonly used in prior inductive KG reasoning models to efficiently construct query-specific subgraphs Zhang et al. (2023). At each GNN layer, we dynamically sample a fixed number of top- k candidate nodes based on their hidden representations, which are updated through query-aware message passing.

Specifically, we employ a trainable scoring function to evaluate the importance of newly encountered nodes and apply top- k selection to retain the most relevant ones. During training, we use Gumbel-softmax to enable differentiable sampling, while during reasoning, hard top- k filtering is applied. This strategy significantly reduces computational overhead while maintaining high-quality subgraph representations for reasoning. Though not the contribution of our work, this sampling method forms an essential part of our pipeline, enabling scalable and query-adaptive multi-hop neighborhood expansion.

3.6. Relation Filtering in Subgraph Construction

To mitigate noise introduced by redundant or semantically irrelevant relations, we introduce a lightweight relation filtering mechanism tailored for inductive KG reasoning. The core idea is to pre-filter relations before subgraph construction, ensuring that only edges semantically aligned with the query relation are preserved.

Given a query triple $(h_q, r_q, ?)$, we compute the cosine similarity between the embedding of the query relation r_q and each relation r_i connected to the query entity h_q . Neighbor edges (r_i, t_i) are retained only if:

$$\cos(r_q, r_i) \geq \tau,$$

where τ is a manually tuned threshold. The resulting filtered neighbor set is:

$$\mathcal{N}(h_q) = \{(r_i, t_i) \mid \cos(r_q, r_i) \geq \tau\}.$$

This pre-filtering step effectively suppresses irrelevant message propagation, producing a more compact and semantically consistent subgraph for subsequent reasoning.

It is worth noting that relation filtering is selectively applied in inductive settings. For large and densely connected transductive KGs, this operation may inadvertently remove informative multi-hop dependencies, leading to performance degradation. In contrast, in large-scale inductive datasets with

diverse and complex relational structures, relation filtering proves beneficial by guiding message propagation toward query-relevant relations and alleviating the impact of redundant connections.

3.7. Algorithmic Details

This section summarizes the complete inference procedure of our proposed GNN-based inductive reasoning framework, integrating QCRA, layer-wise message passing, and the ESAF module. Given a query triple $(e_q, r_q, ?)$, the algorithm constructs a query-specific subgraph, performs multi-layer relational graph encoding with attention-guided aggregation, and iteratively fuses hierarchical representations to produce the final node embeddings for candidate scoring.

The overall process is outlined in Algorithm 1.

4. Experiments

4.1. Experimental Setup

To comprehensively evaluate the effectiveness of our proposed model, we conduct experiments in two widely adopted knowledge graph reasoning settings: (1) Transductive reasoning: All entities are observed during training, while the evaluation is performed on unseen triples within the same graph. (2) Inductive reasoning: The model is required to generalize to entities or structural patterns unseen during training, reflecting realistic scenarios such as evolving knowledge bases. We adopt the commonly used filtered ranking-based metrics, including: Mean Reciprocal Rank (MRR): Measures the average inverse rank of the correct entity. Hits@ k (e.g., Hits@1, Hits@10): Measures the proportion of correct entities ranked in the top- k .

4.2. Datasets

For the transductive setting, we evaluate our approach on five standard benchmarks: 1. **WN18RR** Dettmers et al. (2018): A subset of WordNet designed for relational reasoning, where inverse relations are removed to prevent trivial reasoning. It is widely used to evaluate path-based relational models. 2. **UMLS** Kok and Domingos (2007): A synthetic benchmark focusing on the unseen logical rule composition. It measures how well models can generalize reasoning chains of various lengths and arity. 3. **KINSHIP** Lin et al. (2018): A challenging benchmark containing multi-relational family

Algorithm 1: Inductive Knowledge Graph Reasoning with ESAF Fusion

Input: Query triple $(e_q, r_q, ?)$;Entity and relation embeddings $\mathbf{E}, \mathbf{R}$;GNN layers $\{f_1, \dots, f_L\}$;Subgraph sampler $\mathcal{S}$;ESAF module $\mathcal{F}$;Final scorer $s(\cdot)$ **Output:** Scores over candidate tail entities**Initialization:**Sample initial subgraph $G^{(0)} = \mathcal{S}(e_q)$ centered at e_q Initialize node states $\mathbf{H}^{(0)}$ using query relation r_q Set memory state $\mathbf{M}^{(0)} \leftarrow \mathbf{H}^{(0)}$ **for** $l = 1$ **to** L **do** Sample $G^{(l)} = \mathcal{S}(G^{(l-1)})$; // Expand to l -hop neighborhood $\mathbf{H}^{(l)} \leftarrow f_l(G^{(l)}, \mathbf{H}^{(l-1)}, r_q)$; // GNN with QCRA Align historical memory: $\mathbf{M}_{\text{align}}^{(l-1)} \leftarrow \text{Align}(\mathbf{M}^{(l-1)})$ Fuse states: $\mathbf{F}^{(l)} \leftarrow \mathcal{F}(\mathbf{H}^{(l)}, \mathbf{M}_{\text{align}}^{(l-1)})$ Residual update: $\mathbf{M}^{(l)} \leftarrow \text{LayerNorm}(\mathbf{F}^{(l)} + \mathbf{H}^{(l)})$ **end****Final scoring:** Compute relevance scores for each entity $s_i = W_f \cdot h_i^{(L)}$:**return** $\{s_i\}$

ties, where reasoning requires capturing hierarchical and compositional relationships. 4. **NELL-995** Xiong et al. (2017b): A subgraph extracted from the NELL knowledge base. It includes highly sparse and heterogeneous relational patterns, testing model robustness. 5. **FB15k-237** Toutanova and Chen (2015): A filtered subset of Freebase with symmetric/inverse relations removed, making the inductive setting particularly challenging due to high graph connectivity.

For the inductive setting, we use three benchmark series constructed from WN18RR, FB15k-237, and NELL-995 following the methodology in the previous work Zhang and Yao (2022); Zhang et al. (2023). The detailed statistics are summarized in Table 2.

Table 1: Statistics of the benchmark datasets used in our experiments. $|\mathcal{E}|$ and $|\mathcal{R}|$ denote the number of entities and relations, respectively. $|\mathcal{Q}_{\text{train}}|$, $|\mathcal{Q}_{\text{val}}|$, and $|\mathcal{Q}_{\text{test}}|$ denote the number of training, validation, and test triples.

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	$ \mathcal{Q}_{\text{train}} $	$ \mathcal{Q}_{\text{val}} $	$ \mathcal{Q}_{\text{test}} $
KINSHIP	104	25	8.5k	1.0k	2.0k
UMLS	135	46	5.3k	1.3k	633
WN18RR	40.9k	11	65.1k	3.0k	3.1k
FB15k-237	14.5k	237	204.1k	17.5k	20.4k
NELL-995	74.5k	200	112.2k	543	2.8k

Table 2: Statistics of the inductive benchmark datasets used in our experiments

Dataset	Version	#Rel	Train			Validation			Test		
			#Ent	#Qry	Fact	#Ent	#Qry	Fact	#Ent	#Qry	Fact
FB15k-237	v1	180	1,594	4,245	4,245	1,594	489	4,245	1,093	205	1,993
	v2	200	2,608	9,739	9,739	2,608	1,166	9,739	1,660	478	4,145
	v3	215	3,668	17,986	17,986	3,668	2,194	17,986	2,501	865	7,406
	v4	219	4,707	27,203	27,203	4,707	3,352	27,203	3,051	1,424	11,714
WN18RR	v1	9	2,746	5,410	5,410	2,746	630	5,410	922	188	1,618
	v2	10	6,954	15,262	15,262	6,954	1,838	15,262	2,757	441	4,011
	v3	11	12,078	25,901	25,901	12,078	3,097	25,901	5,084	605	6,327
	v4	9	3,861	7,940	7,940	3,861	934	7,940	7,084	1,429	12,334
NELL-995	v1	14	3,103	4,687	4,687	3,103	414	4,687	225	100	833
	v2	86	2,564	15,262	8,219	8,219	922	8,219	2,086	476	4,586
	v3	142	4,647	16,393	16,393	4,647	1,851	16,393	3,566	809	8,048
	v4	76	2,092	7,546	7,546	2,092	876	7,546	2,795	7,073	731

4.3. Implementation Details

All experiments are conducted on a Linux server running Ubuntu 24.04.1 LTS, equipped with an AMD Ryzen Threadripper PRO 5975WX processor, 256 GB of RAM, and an NVIDIA RTX A6000 GPU with 48 GB of video memory. The models are implemented using PyTorch 1.13.1 with CUDA 11.7. For fair comparison, we adopt consistent training strategies across all models, and the optimal hyperparameter configurations for each dataset are listed in Table 3.

4.4. Performance Comparison on the Transductive Datasets

We comprehensively evaluate the proposed framework against competitive baselines on five public benchmarks under the transductive setting.

Table 3: Hyper-parameter configurations for each dataset.

Dataset	L	K	Emb _d	Batch Size
KINSHIP	5	100	64	10
UMLS	5	100	64	10
WN18RR	8	1000	64	50
FB15k-237	7	2000	48	20
NELL-995	6	2000	64	40

Table 4: Link prediction performance (MRR, Hits@1, Hits@10) on five benchmark datasets under the transductive setting. Best results are in **bold**.

Method	WN18RR			UMLS			KINSHIP			NELL-995			FB15k-237		
	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10
ConvE	0.427	39.2	49.8	0.937	92.2	96.7	0.830	74.0	98.0	0.511	44.6	61.9	0.325	23.7	50.1
SelectE	0.493	45.2	57.4	0.929	87.1	99.8	<u>0.882</u>	<u>81.6</u>	<u>98.5</u>	—	—	—	0.362	26.6	55.4
RNNLogic	0.483	44.6	55.8	0.842	77.2	96.5	0.722	59.8	94.9	0.416	36.3	47.8	0.344	25.2	53.0
RED-GNN	0.533	48.5	62.4	0.964	94.6	99.0	—	—	—	0.543	47.6	<u>65.1</u>	0.374	28.3	55.8
AdaProp	<u>0.562</u>	<u>49.9</u>	<u>67.1</u>	<u>0.969</u>	<u>95.6</u>	99.5	0.711	57.3	96.5	0.554	49.3	65.5	<u>0.417</u>	<u>33.1</u>	<u>58.5</u>
Ours	0.578	52.4	67.8	0.985	97.7	<u>99.6</u>	0.930	88.0	99.4	<u>0.551</u>	<u>48.8</u>	64.8	0.422	33.2	59.8

All comparison methods include ConvE Dettmers et al. (2018), SelectE Zu et al. (2024), RNNLogic Qu et al. (2021), RED-GNN Zhang and Yao (2022), AdaProp Zhang et al. (2023), RuleN Meilicke et al. (2018), DRUM Sadeghian et al. (2019), GraIL Teru et al. (2020), CoMPILE Mai et al. (2021).

Table 4 summarizes the link prediction results of our model and several competitive baselines under the transductive setting across five widely used datasets. Our approach achieves the best overall performance on WN18RR, UMLS, KINSHIP, and FB15k-237, while remaining highly competitive on NELL-995. On WN18RR, our model achieves an MRR of 0.578 and Hits@10 of 67.8%, outperforming the strongest baseline AdaProp (0.562 and 67.1%). This improvement highlights the effectiveness of query-guided relational aggregation in enhancing semantic discrimination during reasoning. On UMLS and KINSHIP, our framework delivers substantial gains, achieving MRR scores of 0.985 and 0.930 respectively—demonstrating its superior capability to capture hierarchical dependencies and logical relations. On FB15k-237, which contains a large number of relation types and dense connections, our model again achieves the highest MRR and Hits@10, showing that the proposed structure-aware fusion effectively mitigates noise propagation and over-smoothing in complex graphs. Although NELL-995 presents greater sparsity and heterogeneity, our model still performs closely to AdaProp, maintaining robust generalization across varied graph structures. Overall, the consistent

improvements across four of the five benchmarks validate the complementary strengths of Query-Conditioned Relational Aggregation (QCRA) and Enhanced Structure-Align Fusion (ESAF). QCRA strengthens the interaction between query semantics and relational context, while ESAF adaptively merges hierarchical representations, yielding a more expressive and stable reasoning process. These findings demonstrate that our framework effectively balances semantic precision and structural depth, achieving strong and reliable performance across diverse transductive reasoning tasks.

4.5. Ablation Study

Table 5: Ablation study results on WN18RR, UMLS, and KINSHIP datasets.

Model Variant	WN18RR			UMLS			KINSHIP			NELL-995			FB15k-237		
	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10
w/o QCRA & ESAF	0.554	50.11	65.56	0.970	95.58	99.13	0.711	57.31	96.46	0.542	47.92	64.19	0.371	29.07	53.04
w/o QCRA	0.552	50.45	64.58	0.970	95.34	99.45	0.864	77.70	98.88	0.541	47.94	63.59	0.375	30.14	52.05
w/o ESAF	0.574	51.98	67.69	0.984	97.87	99.4	0.923	86.73	99.3	0.543	48.19	64.21	0.400	31.63	56.79
Ours (Full Model)	0.578	52.47	67.88	0.985	97.79	99.68	0.930	88.08	99.49	0.551	48.88	64.82	0.422	33.2	59.8

To investigate the contribution of each component in our proposed framework, we conducted ablation experiments on three representative datasets: WN18RR, UMLS, and KINSHIP. The results are summarized in Table 5.

We evaluated three variants of our model:

w/o QCRA & ESAF: Removes both the Query-Conditioned Relational Aggregator and the Enhanced Structure Align Fusion, reverting to a basic GNN reasoning model;

w/o QCRA: Removes only the Query-Conditioned Relational Aggregator, using uniform weights in message passing;

w/o ESAF: Replaces the proposed structure-aware fusion with a standard GRU-based layer aggregation, which sequentially combines representations from different GNN layers without explicit structural alignment.

The results clearly demonstrate that both modules play essential and complementary roles in enhancing reasoning performance. Removing both components (w/o QCRA & ESAF) causes a substantial performance drop across all datasets, confirming that simple graph propagation is insufficient for multi-hop reasoning. The absence of QCRA leads to a notable decline, where message propagation heavily depends on the semantic compatibility between query relations and neighboring structures. Meanwhile, removing ESAF consistently weakens performance, particularly on inductive benchmarks like NELL-995 and FB15k-237, highlighting the necessity of hierarchical representation alignment for robust generalization. Overall, the full

model achieves the highest scores on all datasets, validating that QCRA and ESAF play complementary roles—QCRA enhances semantic selectivity during propagation, while ESAF strengthens cross-layer feature integration for deeper reasoning.

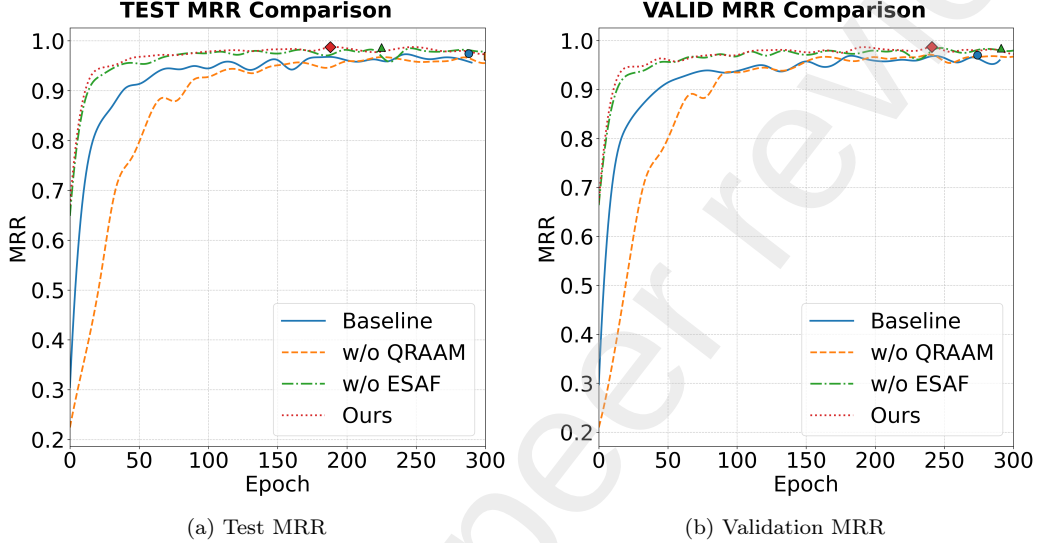


Figure 3: MRR performance across training epochs on the UMLS dataset.

In addition to numerical evaluation, we further visualize the performance dynamics across training in Figure 3, which plots MRR values versus training epochs on UMLS. As shown, our full model not only converges faster but also achieves a significantly higher plateau compared to the ablated variants. This demonstrates that our architectural design not only improves the final performance but also leads to more stable and efficient optimization during training.

4.6. Effect of GNN Depth on Reasoning Performance

To investigate the impact of GNN depth on reasoning performance, we compare our model with two strong baselines—AdaProp and NBFNet—under varying numbers of propagation layers on the WN18RR dataset. As shown in Figure 4, all models benefit to some extent from deeper architectures, as increased depth allows information to propagate over longer paths. However, the magnitude and consistency of performance gains differ notably across models.

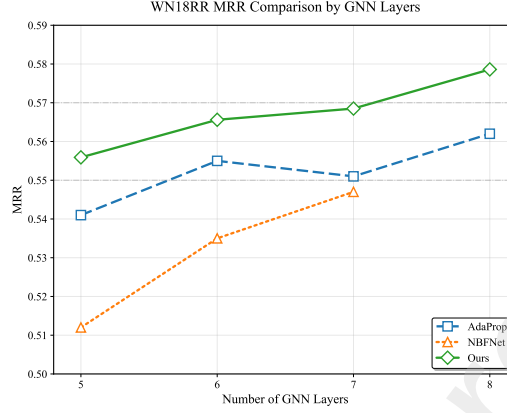


Figure 4: GNN layers comparison.

AdaProp and NBFNet exhibit gradual improvements with increased depth, but their gains tend to plateau or fluctuate at higher layers, suggesting sensitivity to over-smoothing or feature redundancy. In contrast, our model consistently delivers superior performance across all evaluated depths. At each layer setting, it outperforms both baselines, demonstrating greater stability and scalability in deep reasoning scenarios. This performance advantage stems from two key components of our design: the query-guided propagation mechanism, which ensures message relevance by conditioning on query semantics; and the ESAF module, which adaptively integrates multi-layer representations to preserve useful signals while mitigating noise. Together, these components enable our framework to maintain high reasoning accuracy across both shallow and deeper GNN configurations, highlighting its robustness and adaptability to varying graph structures.

4.7. Inference Efficiency Analysis

To evaluate the computational efficiency of our proposed framework, we compare the validation inference time and parameter count of our model against several ablated variants and a strong baseline (AdaProp Zhang et al. (2023)). The results are summarized in Table 6.

Adadrop serves as the baseline model using conventional attention without query awareness or hierarchical fusion. Model 1 replaces our ESAF module with a GRU-based layer aggregation, keeping QCRA unchanged. Model 2 removes QCRA and employs standard attention while retaining ESAF. Model 3 replaces both QCRA and ESAF with traditional mechanisms, ef-

fectively serving as a full ablation. Ours integrates both QCRA and ESAF modules.

As shown in Table 6, our full model achieves the shortest validation inference time across all three datasets, while maintaining a compact parameter scale. When either QCRA or ESAF is removed, the inference time increases moderately, indicating that both modules contribute to computational efficiency. This improvement stems from the design of QCRA and ESAF. QCRA performs query-conditioned attention via efficient key-value projection, reducing redundant edge computation, while ESAF replaces sequential GRU-based fusion with a parallel attention mechanism, enabling lightweight integration of multi-layer features. Together, they streamline message passing and fusion, leading to faster and more stable inference.

Table 6: Validation Inference Efficiency on NELL-995, WN18RR, and FB15k-237.

Model	NELL-995	WN18RR	FB15k-237	Params
Adadrop	26	73	1858	195K
Model 1 (w/o ESAF)	24	75	1899	218K
Model 2 (w/o QCRA)	23	75	1845	221K
Model 3 (w/o ESAF+QCRA)	19	72	2129	102K
Ours (ESAF+QCRA)	18	67	1823	104K

Note: Lower time (unit: second) indicates higher efficiency. Params are abbreviated with "K" (1K = 1000) for readability.

4.8. Performance Comparison on the Inductive Datasets

We further evaluate our framework and its enhanced variant against several inductive reasoning models, including RuleN, DRUM, GraIL, CoMPLE, RED-GNN, and AdaProp, on three widely used inductive benchmarks: WN18RR, FB15k-237, and NELL-995. As summarized in Table 7, our base model (Ours) consistently achieves strong performance across all datasets. In particular, it surpasses AdaProp on WN18RR across all variants, demonstrating the effectiveness of our query-conditioned attention and structure-aware fusion in modeling semantically rich but structurally sparse graphs. However, the limited improvements on FB15k-237(V1-V4) and NELL-995(V2-V4) mainly arise from their data characteristics—large relation vocabularies

but small inductive subgraph sizes, leading to sparse supervision per relation and thus weaker generalization in query-conditioned reasoning.

To address this, we introduce Relation Filtering in Subgraph Construction (Ours+RF), which selectively preserves query-relevant relations during neighborhood expansion. This refinement effectively suppresses relational noise and yields notable improvements on FB15k-237 and NELL-995, confirming its advantage in inductive reasoning scenarios. However, on transductive datasets, such filtering brings limited gains, as their dense connectivity already provides sufficient contextual diversity.

Table 7: Link prediction performance (**MRR**) on three benchmark datasets under the inductive setting. Best results are in **bold**.

Method	WN18RR				FB15k-237				NELL-995			
	V1	V2	V3	V4	V1	V2	V3	V4	V1	V2	V3	V4
RuleN	0.668	0.645	0.368	0.624	0.363	0.433	0.439	0.429	0.615	0.385	0.381	0.333
DRUM	0.666	0.646	0.380	0.627	0.333	0.395	0.402	0.410	0.628	0.365	0.375	0.273
GraIL	0.627	0.625	0.323	0.553	0.279	0.276	0.251	0.227	0.481	0.297	0.322	0.262
CoMPILE	0.577	0.578	0.308	0.548	0.287	0.276	0.262	0.213	0.330	0.248	0.319	0.229
RED-GNN	0.701	0.690	0.427	0.651	0.369	0.469	0.445	0.442	0.637	0.419	0.436	0.363
AdaProp	0.733	0.715	0.474	0.662	0.310	0.471	0.471	0.454	0.644	0.452	0.435	0.366
Ours	0.746	0.742	0.488	0.668	0.383	0.464	0.373	0.436	0.771	0.387	0.390	0.325
Ours+RF	0.743	0.739	0.510	0.656	0.484	0.514	0.494	0.469	0.704	0.504	0.534	0.373

Table 8: Link prediction performance (**Hits@10**) on three benchmark datasets under the inductive setting. Best results are in **bold**.

Method	WN18RR				FB15k-237				NELL-995			
	V1	V2	V3	V4	V1	V2	V3	V4	V1	V2	V3	V4
RuleN	73.0	69.4	40.7	68.1	44.6	59.9	60.0	60.5	76.0	51.4	53.1	48.4
Neural LP	77.2	74.9	47.6	70.6	46.8	58.6	57.1	59.3	87.1	56.4	57.6	53.9
DRUM	77.7	74.7	47.7	70.2	47.4	59.5	57.1	59.3	87.3	54.0	57.7	53.1
GraIL	76.0	77.6	40.9	68.7	42.9	42.4	42.4	38.9	56.5	49.6	51.8	50.6
CoMPILE	74.7	74.3	40.6	67.0	43.9	45.7	44.9	35.8	57.5	44.6	51.5	42.1
NBFNet	82.7	79.9	56.3	70.2	51.7	63.9	58.8	55.9	79.5	63.5	60.6	59.1
RED-GNN	79.9	78.0	52.4	72.1	48.3	62.9	60.3	62.1	86.6	60.1	59.4	55.6
AdaProp	86.6	83.6	62.6	75.5	55.1	65.9	63.7	63.8	88.6	65.2	61.8	60.7
Ours	90.8	89.8	61.1	77.9	51.9	64.9	78.9	62.8	88.6	59.4	54.2	48.6
Ours+RF	91.7	89.8	73.5	81.5	87.5	81.9	87.4	81.7	86.3	75.4	77.9	68.1

4.9. Hits@10 Performance on Inductive Benchmarks

Complementing the MRR analysis, Table 8 presents the Hits@10 results across all inductive benchmarks. The trends are consistent with the previous

findings: our base model maintains strong overall accuracy, particularly on WN18RR, where it consistently demonstrates high retrieval quality across all variants. However, the performance gains are relatively smaller on FB15k-237(V1-V4) and NELL-995(V2-V4), which can be attributed to their large number of relation types and limited inductive subgraph scales. This imbalance results in sparse relational supervision, making it more challenging to capture fine-grained dependencies in inductive reasoning.

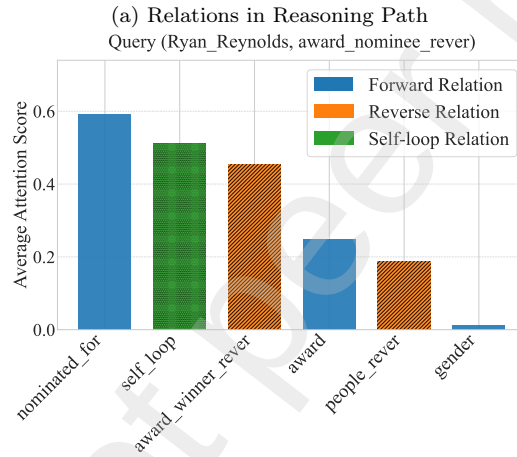
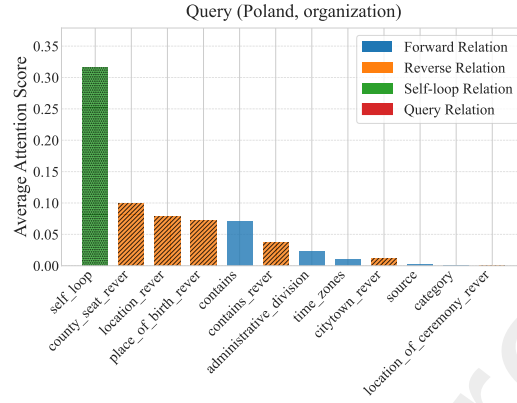
After incorporating Relation Filtering (RF) during subgraph construction, the model achieves substantial and stable improvements. By selectively preserving query-relevant relational contexts, the refined framework produces more focused and reliable predictions. Notably, Ours+RF attains the best results across all datasets and variants, surpassing prior state-of-the-art approaches by a clear margin.

4.10. Visualization of Attention Scores

To better understand how our QCRA facilitates query-aware reasoning, we visualize the average attention scores assigned to relations along reasoning paths for one representative query (Figure 5). For the case with the query (Poland, organization), the model assigns the highest weight to the self-loop relation, which stabilizes entity representation and reduces noise propagation. Among other relations, those semantically connected to geographical or organizational context—such as `county_seat_rever`, `location_rever`, and `place_of_birth_rever`—receive higher scores compared to irrelevant relations like `category` or `location_of_ceremony_rever`, which are almost ignored. This indicates that the model selectively emphasizes relations that align with the query semantics, even in a noisy and heterogeneous subgraph. In general, these visualizations confirm that QCRA effectively integrates query semantics into message propagation, leading to more interpretable and context-sensitive reasoning. The selective emphasis on semantically relevant relations improves the model’s ability to trace reasoning paths that align with the query intent, a key advantage over traditional query-agnostic GNN aggregators.

5. Conclusion

In this work, we propose a GNN-based reasoning framework that explicitly incorporates query relation semantics into message passing and adaptively fuses multi-layer representations. Through the Query-Conditioned Re-



(b) Relations in Reasoning Path

Figure 5: Visualization of attention scores over relations for a representative query.

lational Aggregator (QCRA) and Enhanced Structure-Align Fusion (ESAF) modules, our model effectively enhances both semantic relevance and hierarchical feature utilization. Extensive experiments across five benchmarks demonstrate its consistent and superior performance. Despite these advances, one limitation lies in representing relations as static embeddings throughout reasoning. Although the query relation guides attention, its fixed representation may constrain adaptability to complex or evolving semantics. Future work will explore dynamic, context-aware relation encoders—such as relational prompts or transformer-based mechanisms—to further improve flexibility and semantic alignment in reasoning over knowledge graphs.

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References

- Balazevic, I., Allen, C., Hospedales, T., 2019. TuckER: Tensor factorization for knowledge graph completion, in: Inui, K., Jiang, J., Ng, V., Wan, X. (Eds.), Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China. pp. 5185–5194.
- Bansal, T., Juan, D.C., Ravi, S., McCallum, A., 2019. A2N: Attending to neighbors for knowledge graph inference, in: Korhonen, A., Traum, D., Màrquez, L. (Eds.), Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics.
- Battaglia, P.W., et al., 2018. Relational inductive biases, deep learning, and graph networks *arXiv:1806.01261*.
- Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., Yakhnenko, O., 2013. Translating Embeddings for Modeling Multi-relational Data, in: Neural Information Processing Systems (NIPS), South Lake Tahoe, United States. pp. 1–9. URL: <https://hal.science/hal-00920777>.
- Chen, M., Zhang, W., Zhu, Y., Zhou, H., Yuan, Z., Xu, C., Chen, H., 2022. Meta-knowledge transfer for inductive knowledge graph embedding, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, Association for Computing Machinery. p. 927–937.
- Chen, T., Hong, R., Guo, Y., Hao, S., Hu, B., 2023. Ms²-gnn: Exploring gnn-based multimodal fusion network for depression detection. *IEEE Transactions on Cybernetics* 53, 7749–7759.
- Chen, X., Jia, S., Xiang, Y., 2020. A review: Knowledge reasoning over knowledge graph. *Expert Systems with Applications* 141, 112948. URL: <https://doi.org/10.1016/j.eswa.2020.112948>.

<https://www.sciencedirect.com/science/article/pii/S0957417419306669>,
doi:<https://doi.org/10.1016/j.eswa.2019.112948>.

- Dettmers, T., Minervini, P., Stenetorp, P., Riedel, S., 2018. Convolutional 2d knowledge graph embeddings, AAAI Press.
- Fu, C., Yao, Y., Wu, J., Zhao, W., He, T., Jiang, X., 2023. Multimodal reasoning for nutrition and human health via knowledge graph embedding, in: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 1901–1904. doi:10.1109/BIBM58861.2023.10385745.
- Ji, S., Pan, S., Cambria, E., Marttinen, P., Yu, P.S., 2022. A survey on knowledge graphs: Representation, acquisition, and applications. IEEE Transactions on Neural Networks and Learning Systems 33, 494–514. doi:10.1109/TNNLS.2021.3070843.
- Kipf, T.N., Welling, M., 2017. Semi-supervised classification with graph convolutional networks, in: International Conference on Learning Representations. URL: <https://openreview.net/forum?id=SJU4ayYgl>.
- Kok, S., Domingos, P., 2007. Statistical predicate invention, in: Proceedings of the 24th International Conference on Machine Learning, p. 433–440.
- Li, J., Luo, X., Lu, G., 2024. Gs-cbr-kbqa: Graph-structured case-based reasoning for knowledge base question answering. Expert Systems with Applications 257, 125090.
- Li, W., Peng, R., Li, Z., 2023. Knowledge graph completion by jointly learning structural features and soft logical rules. IEEE Transactions on Knowledge and Data Engineering 35, 2724–2735. doi:10.1109/TKDE.2021.3108224.
- Liao, N., Mo, D., Luo, S., Li, X., Yin, P., 2022. Scara: scalable graph neural networks with feature-oriented optimization. Proc. VLDB Endow. 15, 3240–3248. URL: <https://doi.org/10.14778/3551793.3551866>, doi:10.14778/3551793.3551866.
- Lin, X.V., Socher, R., Xiong, C., 2018. Multi-hop knowledge graph reasoning with reward shaping, in: Riloff, E., Chiang, D., Hockenmaier, J., Tsujii, J. (Eds.), Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, Brussels, Belgium. pp. 3243–3253.

- Lin, Y., Liu, Z., Sun, M., Liu, Y., Zhu, X., 2015. Learning entity and relation embeddings for knowledge graph completion, in: Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence, p. 2181–2187.
- Liu, Q., Zhang, H., Liu, Z., 2021a. Simplifying graph convolutional networks as matrix factorization, in: U, L.H., Spaniol, M., Sakurai, Y., Chen, J. (Eds.), Web and Big Data, Springer International Publishing, Cham. pp. 35–43.
- Liu, S., Grau, B.C., Horrocks, I., Kostylev, E.V., 2021b. Indigo: Gnn-based inductive knowledge graph completion using pair-wise encoding, in: Proceedings of the 35th International Conference on Neural Information Processing Systems, Curran Associates Inc., Red Hook, NY, USA.
- Mai, S., Zheng, S., Yang, Y., Hu, H., 2021. Communicative message passing for inductive relation reasoning, in: Proceedings of the AAAI conference on artificial intelligence, pp. 4294–4302.
- Meilicke, C., Fink, M., Wang, Y., Ruffinelli, D., Gemulla, R., Stuckenschmidt, H., 2018. Fine-grained evaluation of rule- and embedding-based systems for knowledge graph completion, in: The Semantic Web – ISWC 2018: 17th International Semantic Web Conference, Monterey, CA, USA, October 8–12, 2018, Proceedings, Part I, Springer-Verlag. p. 3–20.
- Qiu, L., Yang, N., Maolimamuti, M., 2017. Chinese-uyghur-english semantic search based on the knowledge graphs, in: 2017 IEEE International Conference on Computational Science and Engineering (CSE) and IEEE International Conference on Embedded and Ubiquitous Computing (EUC), pp. 879–882. doi:10.1109/CSE-EUC.2017.177.
- Qu, M., Chen, J.K., Xhonneux, L.P.A.C., Bengio, Y., Tang, J., 2021. Rnn-logic: Learning logic rules for reasoning on knowledge graphs, in: ICLR.
- Sadeghian, A., Armandpour, M., Ding, P., Wang, D.Z., 2019. Drum: End-to-end differentiable rule mining on knowledge graphs. *Advances in neural information processing systems* 32.
- Schlichtkrull, M., Kipf, T.N., Bloem, P., van nbsp, 2018. Modeling relational data with graph convolutional networks, in: The Semantic Web: 15th International Conference, ESWC 2018, Heraklion, Crete, Greece, June 3–7, 2018, Proceedings, p. 593–607.

- Sun, Z., Deng, Z., Nie, J.Y., Tang, J., 2018. Rotate: Knowledge graph embedding by relational rotation in complex space. ArXiv abs/1902.10197. URL: <https://api.semanticscholar.org/CorpusID:67855617>.
- Tang, M., Wu, D., Zhang, S., Gao, W., 2024. Epan-serec: Expertise preference-aware networks for software expert recommendations with knowledge graph. *Expert Systems with Applications* 244, 122985.
- Teru, K.K., Denis, E.G., Hamilton, W.L., 2020. Inductive relation prediction by subgraph reasoning, in: *Proceedings of the 37th International Conference on Machine Learning, JMLR.org*.
- Toutanova, K., Chen, D., 2015. Observed versus latent features for knowledge base and text inference, in: Allauzen, A., Grefenstette, E., Hermann, K.M., Larochelle, H., Yih, S.W.t. (Eds.), *Proceedings of the 3rd Workshop on Continuous Vector Space Models and their Compositionality*, Beijing, China. pp. 57–66.
- Trouillon, T., Welbl, J., Riedel, S., Gaussier, É., Bouchard, G., 2016. Complex embeddings for simple link prediction. ArXiv abs/1606.06357. URL: <https://api.semanticscholar.org/CorpusID:15150247>.
- Vashishth, S., Sanyal, S., Nitin, V., Agrawal, N., Talukdar, P., 2020a. Interact: Improving convolution-based knowledge graph embeddings by increasing feature interactions, in: *Proceedings of the AAAI conference on artificial intelligence*, pp. 3009–3016.
- Vashishth, S., Sanyal, S., Nitin, V., Talukdar, P., 2020b. Composition-based multi-relational graph convolutional networks, in: *International Conference on Learning Representations*.
- Wang, G., Ying, R., Huang, J., Leskovec, J., 2021a. Multi-hop attention graph neural networks, in: Zhou, Z.H. (Ed.), *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, IJCAI-21, International Joint Conferences on Artificial Intelligence Organization*. pp. 3089–3096. URL: <https://doi.org/10.24963/ijcai.2021/425>, doi:10.24963/ijcai.2021/425. main Track.
- Wang, K., Liu, Y., Lin, D., Sheng, M., 2021b. Hyperbolic geometry is not necessary: Lightweight Euclidean-based models for low-dimensional knowledge graph embeddings, in: Moens, M.F., Huang,

- X., Specia, L., Yih, S.W.t. (Eds.), Findings of the Association for Computational Linguistics: EMNLP 2021, Association for Computational Linguistics, Punta Cana, Dominican Republic. pp. 464–474. URL: <https://aclanthology.org/2021.findings-emnlp.42/>, doi:10.18653/v1/2021.findings-emnlp.42.
- Xiong, C., Power, R., Callan, J., 2017a. Explicit semantic ranking for academic search via knowledge graph embedding, in: Proceedings of the 26th International Conference on World Wide Web, p. 1271–1279.
- Xiong, W., Hoang, T., Wang, W.Y., 2017b. DeepPath: A reinforcement learning method for knowledge graph reasoning, in: Palmer, M., Hwa, R., Riedel, S. (Eds.), Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, Copenhagen, Denmark. pp. 564–573.
- Yang, B., tau Yih, W., He, X., Gao, J., Deng, L., 2014. Embedding entities and relations for learning and inference in knowledge bases, in: International Conference on Learning Representations. URL: <https://api.semanticscholar.org/CorpusID:2768038>.
- Zhang, Y., Yao, Q., 2022. Knowledge graph reasoning with relational di-graph, in: Proceedings of the ACM Web Conference 2022, p. 912–924.
- Zhang, Y., Zhou, Z., Yao, Q., Chu, X., Han, B., 2023. Adaprop: Learning adaptive propagation for graph neural network based knowledge graph reasoning, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, p. 3446–3457.
- Zhao, Y., Qi, J., Liu, Q., Zhang, R., 2021. WgcN: Graph convolutional networks with weighted structural features, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, p. 624–633.
- Zhu, Z., Yuan, X., Galkin, M., Xhonneux, L.P., Zhang, M., Gazeau, M., Tang, J., 2023. A * net: A scalable path-based reasoning approach for knowledge graphs, in: Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., Levine, S. (Eds.), Advances in Neural Information Processing Systems, pp. 59323–59336.

Zhu, Z., Zhang, Z., Khonneux, L.P., Tang, J., 2021. Neural bellman-ford networks: a general graph neural network framework for link prediction, in: Proceedings of the 35th International Conference on Neural Information Processing Systems.

Zu, L., Lin, L., Fu, S., Guo, F., Wu, J., 2024. Selecte: Multi-scale adaptive selection network for knowledge graph representation learning. Knowledge-Based Systems 291, 111554.